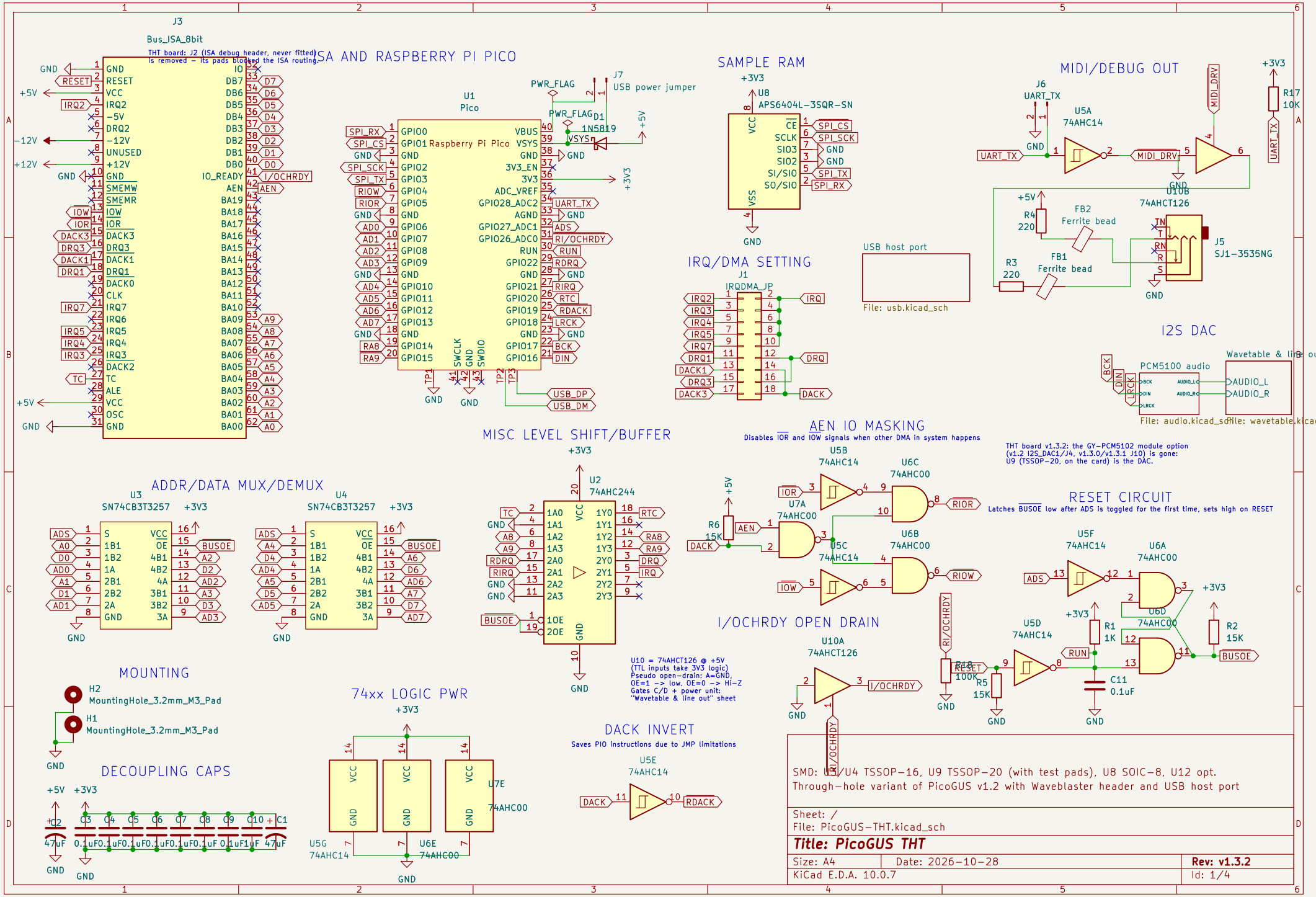
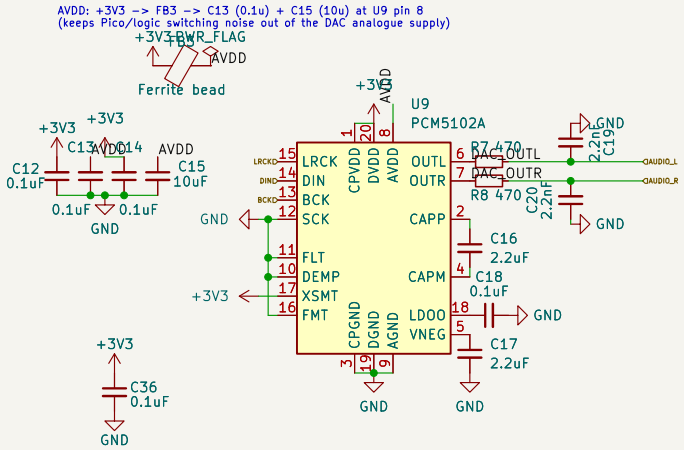

ISA AND RASPBERRY PI PICO
SAMPLE RAM
MIDI/DEBUG OUT
IRQ/DMA SETTING
USB host port
I2S DAC
MISC LEVEL SHIFT/BUFFER
AEN IO MASKING
ADDR/DATA MUX/DEMUX
RESET CIRCUIT
I/OCHRDY OPEN DRAIN
MOUNTING
74xx LOGIC PWR
DECOUPLING CAPS
DACK INVERT
J3
Bus_ISA_8bit
THT board: J2 (ISA debug header, never fitted) is removed — its pads blocked the ISA routing.
U1
Pico
Raspberry Pi Pico
J7
USB power jumper
PWR_FLAG
D1
1N5819
U8
APS6404L-3SQR-SN
J6
UART_TX
U5A
74AHC14
U10B
74AHCT126
J5
SJ1-3535NG
FB1
Ferrite bead
FB2
Ferrite bead
File: usb.kicad_sch
J1
IRQDMA_JP
PCM5100 audio
Wavetable & line out
AUDIO_L
AUDIO_R
File: audio.kicad_sch
File: wavetable.kicad_sch
U2
74AHC244
U5B
74AHC14
U6C
74AHC00
U7A
74AHC00
U6B
74AHC00
U5C
74AHC14
THT board v1.3.2: the GY-PCM5102 module option (v1.2 I2S_DAC1/J4, v1.3.0/v1.3.1 J10) is gone: U9 (TSSOP-20, on the card) is the DAC.
Disables IOR and IOW signals when other DMA in system happens
Latches BUSOE low after ADS is toggled for the first time, sets high on RESET
U3
SN74CB3T3257
U4
SN74CB3T3257
U5F
74AHC14
U6A
74AHC00
U6D
74AHC00
U5D
74AHC14
U10A
74AHCT126
U10 = 74AHCT126 @ +5V (TTL inputs take 3V3 logic) Pseudo open-drain: A=GND, OE=1 -> low, OE=0 -> Hi-Z Gates C/D + power unit: "Wavetable & line out" sheet
U5E
74AHC14
U5G
74AHC14
U6E
74AHC00
U7E
74AHC00
H2
MountingHole_3.2mm_M3_Pad
H1
MountingHole_3.2mm_M3_Pad
SMD: U1/U4 TSSOP-16, U9 TSSOP-20 (with test pads), U8 SOIC-8, U12 opt.
Through-hole variant of PicoGUS v1.2 with Waveblaster header and USB host port
Sheet: /
File: PicoGUS-THT.kicad_sch
Title: PicoGUS THT
Size: A4
Date: 2026-10-28
Rev: v1.3.2
KiCad E.D.A. 10.0.7
Id: 1/4




U9 PCM5102A (TSSOP-20; PCM5100A/5101A fit too) soldered to the card, with one test pad per pin

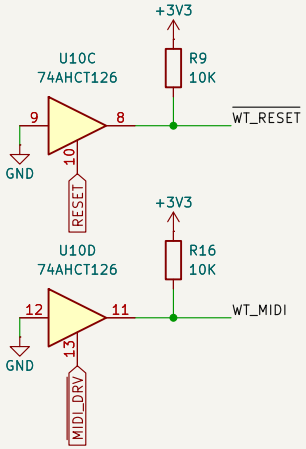
Sheet: /PCM5100 audio/
File: audio.kicad_sch

**Title: PicoGUS THT – PCM510x DAC**

| Size: A4 | Date: 2026-10-23 | **Rev: v1.3.2** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 2/4 |

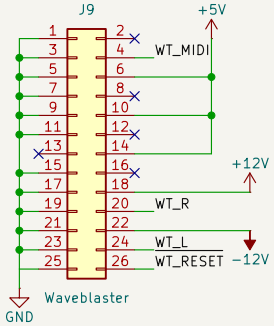
## WT CONTROL (U10 = 74AHCT126 @ +5V)

Both outputs are pseudo open-drain (A=GND, OE drives) with pull-ups to +3V3, so the daughterboard never sees 5 V on /RESET or MIDI (some DBs have 3.3 V inputs).
/RESET: low while ISA RESET is high.  MIDI: same data as the MIDI-out jack (MPU-401 UART).



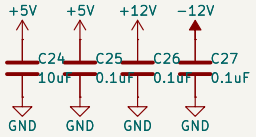

## WAVEBLASTER HEADER

2x13 male header, 2.54 mm. Daughterboard (female socket) sits over the card; footprint has the standard standoff holes.

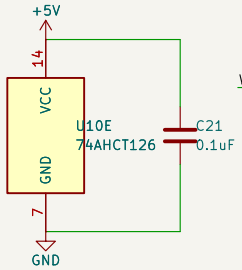


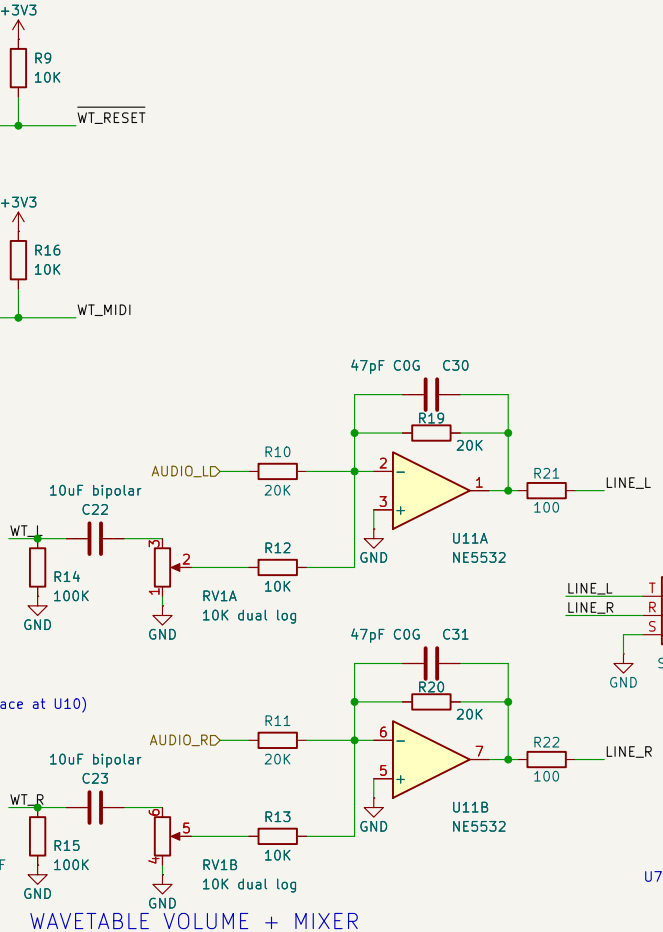

DGND 1–11, AGND 15–25: both to the ground plane

## WT SUPPLY DECOUPLING (place at J9)



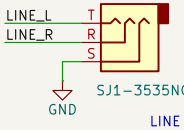

## U10 POWER (+5V) + DECOUPLING (place at U10)



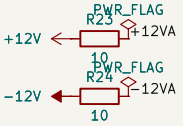




## WAVETABLE VOLUME + MIXER

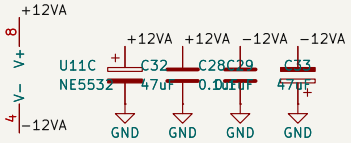
RV1 (dual 10k log) sets the WT level: pins 3/6 = CW end (signal), 1/4 = CCW end (GND).
U11 is an inverting summer: PCM 20k in / 20k fb (−unity, −0.2 dB), WT 10k in (x2 = +6 dB at RV1 max; DB outputs are ~1–1.2 Vrms vs 2.1 Vrms PCM). The WT knob does not change the PCM level. C22/C23 (bipolar) block DC from the DB, R14/R15 keep them discharged when no DB is fitted. Output: 100R, DC-coupled like v1.2.



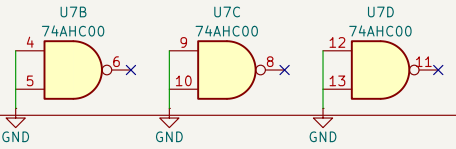

LINE OUT (green)




U11 SUPPLY: ISA +/−12 V via R23/R24 (10R) + C32/C33 (47u), C2




## U7 SPARE GATES (74AHC00)

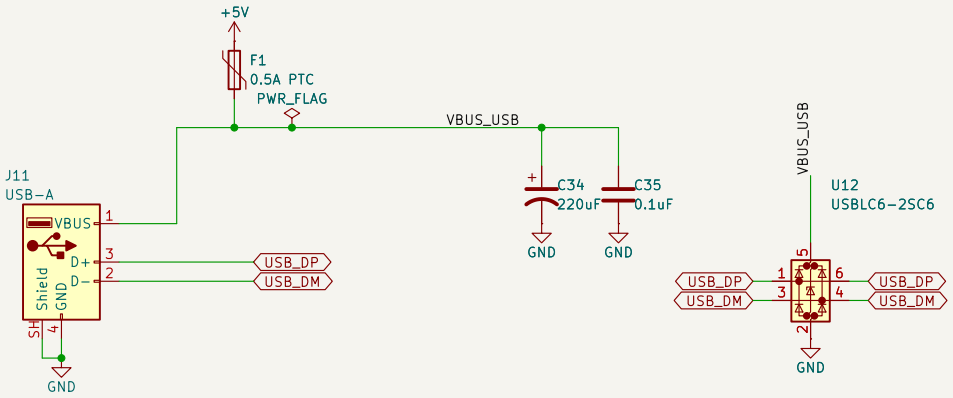



Derived from the hand386 ISA adapter mixer and the PicoGUS 2.0 WT header
Waveblaster (wavetable) daughterboard header, analog WT volume and op-amp mixer

Sheet: /Wavetable & line out/
File: wavetable.kicad_sch

**Title: PicoGUS THT — Waveblaster header & line-out mixer**

| Size: A4 | Date: 2026-10-23 | **Rev: v1.3.2** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 3/4 |




## USB HOST PORT (rev B)

Data straight from the Pico's USB test points TP3 (D+) / TP2 (D−) (U1, sheet 1), behind the 27R series resistors on the Pico module. The Pico must be soldered FLAT on the board (castellated edge or through-hole pins, no header) so TP1–TP3 can be soldered from the back of the card.
With the Pico on headers or in a socket the TP pads do not reach the card: wire TP1–TP3 to the holes, or use a micro-USB OTG adapter on the Pico (with jumper J7) as on v1.3.0. Never use both at once.
VBUS from ISA +5V via F1 (resettable, 0.5 A hold), C34 bulk. The Pico VBUS pin is not used for this port.
U12 (ESD array, SOT-23-6) is optional.

J11: USB-A receptacle (Molex 67643) at the bracket
USB host port (mouse, joystick, storage) on the Pico USB test points TP2/TP3

Sheet: /USB host port/
File: usb.kicad_sch

**Title: PicoGUS THT – USB host port**

| Size: A4 | Date: 2026-10-23 | **Rev: v1.3.2** |
|---|---|---|
| KiCad E.D.A. 10.0.7 | | Id: 4/4 |